**MAT 3103: Computational Statistics and Probability** **Semester: Fall 2019-20**




# Chapter 9: Correlation and Regression

*Ashfia Tasnim Munia, Md. Mortuza Ahmmed, Dr. Mahfuza Khatun*

## 9.1 Correlation:

Pearson's correlation coefficient is the covariance of the two variables divided by the product of their standard deviations.

**For population**,

Pearson's correlation coefficient when applied to a population is commonly represented by the Greek letter $\rho$ (rho) and may be referred to as the population correlation coefficient or the population Pearson correlation coefficient. The formula for $\rho$ is:

$\rho = \frac{COV(X,Y)}{\sigma_X \sigma_Y}$,

where:

$COV$ is the covariance

$\sigma_X$ & $\sigma_Y$ is the standard deviation of $X$ & $Y$ respectively.

**For sample**,

Pearson's correlation coefficient when applied to a sample is commonly represented by the letter r and may be referred to as the sample correlation coefficient or the sample Pearson correlation coefficient. We can obtain a formula for $r$ by substituting estimates of the covariances and variances based on a sample into the formula above. Let $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ be $n$ pairs of observations of variables $X$ and $Y$ observed from $n$ sample points. The linear relationship of $X$ and $Y$ is called simple correlation. The degree of linear

relationship of $X$ and $Y$ is estimated by a quantity $r$, where

$$\begin{aligned} r &= \frac{cov(x,y)}{\sqrt{v(x)v(y)}} \\ &= \frac{\frac{1}{n}\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\frac{1}{n}\sum(x-\bar{x})^2\frac{1}{n}\sum(y-\bar{y})^2}} \\ &= \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{[\sum x^2 - \frac{(\sum x)^2}{n}][\sum y^2 - \frac{(\sum y)^2}{n}]}} \\ &= \frac{sp(xy)}{\sqrt{ss(x)ss(y)}} \end{aligned}$$

This $r$ is called simple correlation coefficient of $x$ and $y$. It measures the strength of linear relationship of $x$ and $y$.

## Example 1:

Given,

| | | | |
|---|---|---|---|
| x | 1 | 2 | 3 |
| y | 3 | 6 | 9 |

Calculate correlation coefficient $x$ and $y$. Also comment according to the result.

**Solution:**

| x | y | xy | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 1 | 3 | 3 | 1 | 9 |
| 2 | 6 | 12 | 4 | 36 |
| 3 | 9 | 27 | 9 | 81 |
| $\sum x$ =6 | $\sum y$ =18 | $\sum xy$ =42 | $\sum x^2$ =14 | $\sum y^2$ =126 |

Here, $n = 3$.

$$\begin{aligned} ss(x) &= \sum x^2 - \frac{(\sum x)^2}{n} = 14 - \frac{(6)^2}{3} = 2 \\ ss(y) &= \sum y^2 - \frac{(\sum y)^2}{n} = 126 - \frac{(18)^2}{3} = 18 \\ sp(xy) &= \sum xy - \frac{\sum x \sum y}{n} = 42 - \frac{6 \times 18}{3} = 6 \\ r &= \frac{sp(xy)}{\sqrt{ss(x)ss(y)}} = \frac{6}{\sqrt{2 \times 18}} = 1 \end{aligned}$$

$r = 1$ indicates that $x$ and $y$ are perfectly and positively correlated.

## Example 2:

Given,

| x | 1 | 2 | 3 |
|---|---|---|---|
| y | 9 | 6 | 3 |

Calculate correlation coefficient $x$ and $y$. Also comment according to the result.

**Solution:**

| x | y | xy | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 1 | 9 | 9 | 1 | 81 |
| 2 | 6 | 12 | 4 | 36 |
| 3 | 3 | 9 | 9 | 9 |
| $\sum x$ =6 | $\sum y$ =18 | $\sum xy$ =30 | $\sum x^2$ =14 | $\sum y^2$ =126 |

Here, $n = 3$.

$$ss(x) = \sum x^2 - \frac{(\sum x)^2}{n} = 14 - \frac{(6)^2}{3} = 2$$
$$ss(y) = \sum y^2 - \frac{(\sum y)^2}{n} = 126 - \frac{(18)^2}{3} = 18$$
$$sp(xy) = \sum xy - \frac{\sum x \sum y}{n} = 30 - \frac{6 \times 18}{3} = -6$$
$$r = \frac{sp(xy)}{\sqrt{ss(x)ss(y)}} = \frac{-6}{\sqrt{2 \times 18}} = -1$$

$r = -1$ indicates that $x$ and $y$ are perfectly and negatively correlated.

## Example 3:

Given,

| x | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| y | 3 | 6 | 9 | 9 | 6 | 3 |

Calculate correlation coefficient $x$ and $y$. Also comment according to the result.

**Solution:**

| x | y | xy | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 1 | 3 | 3 | 1 | 9 |
| 2 | 6 | 12 | 4 | 36 |
| 3 | 9 | 27 | 9 | 81 |
| 4 | 9 | 36 | 16 | 81 |
| 5 | 6 | 30 | 25 | 36 |
| 6 | 3 | 18 | 36 | 9 |
| $\sum x$ =21 | $\sum y$ =36 | $\sum xy$ =126 | $\sum x^2$ =91 | $\sum y^2$ =252 |

Here, $n = 6$.

$$sp(xy) = \sum xy - \frac{\sum x \sum y}{n} = 126 - \frac{21 \times 36}{6} = 0$$
$$r = \frac{sp(xy)}{\sqrt{ss(x)ss(y)}} = \frac{0}{\sqrt{ss(x)ss(y)}} = 0$$

$r = 0$ indicates that, there is no correlation between $x$ and $y$.

## Example 4:

Given the number of ever born children ($y$) to different mothers having different levels of education ($x$ in completed years of schooling):

| x | 8 | 4 | 5 | 10 | 12 | 8 | 5 | 10 | 0 | 6 | 8 | 5 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| y | 2 | 6 | 5 | 3 | 1 | 2 | 5 | 2 | 7 | 3 | 4 | 2 | 5 |

(a)Calculate correlation coefficient $x$ and $y$.
(b)Test the significance of correlation.

**Solution:**

| x | y | xy | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 8 | 2 | 16 | 64 | 4 |
| 4 | 6 | 24 | 16 | 36 |
| 5 | 5 | 25 | 25 | 25 |
| 10 | 3 | 30 | 100 | 9 |
| 12 | 1 | 12 | 144 | 1 |
| 8 | 2 | 16 | 64 | 4 |
| 5 | 5 | 25 | 25 | 25 |
| 10 | 2 | 20 | 100 | 4 |
| 0 | 7 | 0 | 0 | 49 |
| 6 | 3 | 18 | 36 | 9 |
| 8 | 4 | 32 | 64 | 16 |
| 5 | 2 | 10 | 25 | 4 |
| 0 | 5 | 0 | 0 | 25 |
| $\sum x$ =81 | $\sum y$ =47 | $\sum xy$ =228 | $\sum x^2$ =663 | $\sum y^2$ =211 |

(a) Here, $n = 13$.

$$ss(x) = \sum x^2 - \frac{(\sum x)^2}{n} = 663 - \frac{(81)^2}{13} = 158.31$$
$$ss(y) = \sum y^2 - \frac{(\sum y)^2}{n} = 211 - \frac{(47)^2}{13} = 41.08$$
$$sp(xy) = \sum xy - \frac{\sum x \sum y}{n} = 228 - \frac{81 \times 47}{13} = -64.85$$
$$r = \frac{sp(xy)}{\sqrt{ss(x)ss(y)}} = \frac{-64.85}{\sqrt{158.31 \times 41.08}} = -0.80$$

$r = -0.80$ indicates that $x$ and $y$ are strongly negatively correlated.

(b) We need to test the hypothesis,

Null hypothesis, $H_0 : \rho = 0$ or, The correlation is not significant

Alternative hypothesis, $H_1 : \rho \neq 0$ or, The correlation is significant.

The test statistic is, $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.80\sqrt{13-2}}{\sqrt{1-(-0.80)^2}} = -4.43$

$t_{cal} = |t| = 4.43$ and $t_{tab} = t_{0.05,n-2} = t_{0.05,13-2} = t_{0.05,11} = 2.201$

$t_{cal} > t_{tab}$ so, $H_0$ is rejected and the correlation is significant.

## 9.2 Regression:

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationships among variables. It includes many techniques for modeling and analyzing several variables, when the focus is on the relationship between a dependent variable and one or more independent variables (or 'predictors'). More specifically, regression analysis helps one understand how the typical value of the dependent variable (or 'criterion variable') changes when any one of the independent variables is varied, while the other independent variables are held fixed.

In statistics, **linear regression** is a linear approach to modeling the relationship between a scalar response (or dependent variable) and one or more explanatory variables (or independent variables). The case of one explanatory variable is called **simple linear regression**. For more than one explanatory variable, the process is called **multiple linear regression**. This term is distinct from **multivariate linear regression**, where multiple correlated dependent variables are predicted, rather than a single scalar variable.

Linear regression is a technique to formulate a mathematical function of dependent variable y based on independent variable x so that for any value of $x$ the value of $y$ can be estimated. The linear regression model is given by, $y = \alpha + \beta x + \epsilon$,

where $\alpha$=The value of $y$ when $x = 0$

$\beta$= Regression coefficient. It measures the rate of change of $y$ for unit change in $x$.

$\epsilon$=Random error. It is used in the model to measure the influences of other variables which are not included in the model.

**Simple linear regression model:** $y = \alpha + \beta x + \epsilon$

**Multiple linear regression model:** $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots \beta_k x_k + \epsilon$,

where, $y = n \times 1$ matrix

$\beta = k \times 1$ matrix

$\epsilon = n \times 1$ matrix

$x = n \times k$ matrix

**Fitted model of simple linear regression:** The problem is to fit the regression equation in such a way that the sum of squares due to error is minimum. Let the fitted model be

$y = a + bx$,

where $a$ is the estimate of $\alpha$ and $b$ is the estimate of $\beta$. Here

$a = \bar{y} - b\bar{x}$ and $b = \frac{SP(xy)}{SS(x)}$ $[ss(x) = \sum x^2 - \frac{(\sum x)^2}{n}$, $sp(xy) = \sum xy - \frac{\sum x \sum y}{n}]$

## Example 5:

Given the number of ever born children ($y$) to different mothers having different levels of education ($x$ in completed years of schooling):

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| x | 8 | 4 | 5 | 10 | 12 | 8 | 5 | 10 | 0 | 6 | 8 | 5 | 0 |
| y | 2 | 6 | 5 | 3 | 1 | 2 | 5 | 2 | 7 | 3 | 4 | 2 | 5 |

(a)Fit a regression line of y on x and estimate the number of children of a mother who complete 14 years of schooling.
(b)Test the significance of regression.

**Solution:**

| x | y | xy | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 8 | 2 | 16 | 64 | 4 |
| 4 | 6 | 24 | 16 | 36 |
| 5 | 5 | 25 | 25 | 25 |
| 10 | 3 | 30 | 100 | 9 |
| 12 | 1 | 12 | 144 | 1 |
| 8 | 2 | 16 | 64 | 4 |
| 5 | 5 | 25 | 25 | 25 |
| 10 | 2 | 20 | 100 | 4 |
| 0 | 7 | 0 | 0 | 49 |
| 6 | 3 | 18 | 36 | 9 |
| 8 | 4 | 32 | 64 | 16 |
| 5 | 2 | 10 | 25 | 4 |
| 0 | 5 | 0 | 0 | 25 |
| $\sum x$ =81 | $\sum y$ =47 | $\sum xy$ =228 | $\sum x^2$ =663 | $\sum y^2$ =211 |

Let the fitted line be $\hat{y} = a + bx$

**(a)** Here, $n = 13$.

$$ss(x) = \sum x^2 - \frac{(\sum x)^2}{n} = 663 - \frac{(81)^2}{13} = 158.31$$
$$ss(y) = \sum y^2 - \frac{(\sum y)^2}{n} = 211 - \frac{(47)^2}{13} = 41.08$$
$$sp(xy) = \sum xy - \frac{\sum x \sum y}{n} = 228 - \frac{81 \times 47}{13} = -64.85$$
$$\bar{x} = \frac{\sum x}{n} = 81/13 = 6.23$$
$$\bar{y} = \frac{\sum y}{n} = 47/13 = 3.62$$
$$b = \frac{sp(xy)}{ss(x)} = \frac{-64.85}{158.31} = -0.41$$
$$a = \bar{y} - b\bar{x} = 3.62 - (-0.41)6.23 = 6.17$$

Therefore the fitted line is, $\hat{y} = 6.17 - 0.41x$

When, $x = 14$, then $\hat{y} = 6.17 - 0.41 \times 14 = 0.49$.

**(b)** We need to test the hypothesis,

$H_0 : \beta = 0$, or the regression is not significant

$H_1 : \beta \neq 0$, or the regression is significant

Test statistic, $t = \frac{b}{\sqrt{\frac{s^2}{SS(x)}}}$.....(1)

$s^2 = \frac{1}{n-2}[SS(y) - bSP(xy)] = \frac{1}{11}[41.08 - (-0.41)(-64.85)] = 1.317$

From equation (1), $t = \frac{-0.41}{\sqrt{\frac{1.317}{41.08}}} = -4.50$

$t_{cal} = |t| = 4.50$ & $t_{tab} = t_{0.05,n-2} = t_{0.05,11} = 2.201$

$t_{cal} > t_{tab}$, so $H_0$ is rejected and we can conclude that the regression is significant.

## 9.3 MATLAB code for correlation and regression:

**Correlation coefficient:** Create X matrix where all of the variables are included.
[r,p]=corrcoef(X)

**Simple linear regression:** Create y (include only response variable) and x (include only explanatory variable).
regress(y, x)

**Multiple linear regression:** Create X matrix where only explanatory variables (independent variable) are included.
regress(y, X)

## *t* Table

| cum. prob | $t_{.50}$ | $t_{.75}$ | $t_{.80}$ | $t_{.85}$ | $t_{.90}$ | $t_{.95}$ | $t_{.975}$ | $t_{.99}$ | $t_{.995}$ | $t_{.999}$ | $t_{.9995}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **one-tail** | **0.50** | **0.25** | **0.20** | **0.15** | **0.10** | **0.05** | **0.025** | **0.01** | **0.005** | **0.001** | **0.0005** |
| **two-tails** | **1.00** | **0.50** | **0.40** | **0.30** | **0.20** | **0.10** | **0.05** | **0.02** | **0.01** | **0.002** | **0.001** |
| **df** | | | | | | | | | | | |
| 1 | 0.000 | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 318.31 | 636.62 |
| 2 | 0.000 | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 22.327 | 31.599 |
| 3 | 0.000 | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 10.215 | 12.924 |
| 4 | 0.000 | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 7.173 | 8.610 |
| 5 | 0.000 | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5.893 | 6.869 |
| 6 | 0.000 | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 5.208 | 5.959 |
| 7 | 0.000 | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.785 | 5.408 |
| 8 | 0.000 | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 4.501 | 5.041 |
| 9 | 0.000 | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 4.297 | 4.781 |
| 10 | 0.000 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 4.144 | 4.587 |
| 11 | 0.000 | 0.697 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 4.025 | 4.437 |
| 12 | 0.000 | 0.695 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.930 | 4.318 |
| 13 | 0.000 | 0.694 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.852 | 4.221 |
| 14 | 0.000 | 0.692 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.787 | 4.140 |
| 15 | 0.000 | 0.691 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.733 | 4.073 |
| 16 | 0.000 | 0.690 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.686 | 4.015 |
| 17 | 0.000 | 0.689 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.646 | 3.965 |
| 18 | 0.000 | 0.688 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 | 3.922 |
| 19 | 0.000 | 0.688 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 | 3.883 |
| 20 | 0.000 | 0.687 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 | 3.850 |
| 21 | 0.000 | 0.686 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 | 3.819 |
| 22 | 0.000 | 0.686 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.505 | 3.792 |
| 23 | 0.000 | 0.685 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.485 | 3.768 |
| 24 | 0.000 | 0.685 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.467 | 3.745 |
| 25 | 0.000 | 0.684 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.450 | 3.725 |
| 26 | 0.000 | 0.684 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.435 | 3.707 |
| 27 | 0.000 | 0.684 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.421 | 3.690 |
| 28 | 0.000 | 0.683 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.408 | 3.674 |
| 29 | 0.000 | 0.683 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.396 | 3.659 |
| 30 | 0.000 | 0.683 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.385 | 3.646 |
| 40 | 0.000 | 0.681 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 3.307 | 3.551 |
| 60 | 0.000 | 0.679 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 3.232 | 3.460 |
| 80 | 0.000 | 0.678 | 0.846 | 1.043 | 1.292 | 1.664 | 1.990 | 2.374 | 2.639 | 3.195 | 3.416 |
| 100 | 0.000 | 0.677 | 0.845 | 1.042 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 3.174 | 3.390 |
| 1000 | 0.000 | 0.675 | 0.842 | 1.037 | 1.282 | 1.646 | 1.962 | 2.330 | 2.581 | 3.098 | 3.300 |
| ***z*** | 0.000 | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 |
| | 0% | 50% | 60% | 70% | 80% | 90% | 95% | 98% | 99% | 99.8% | 99.9% |
| | **Confidence Level** | | | | | | | | | | |

t-table.xls 7/14/2007

Figure 9.1: t-table